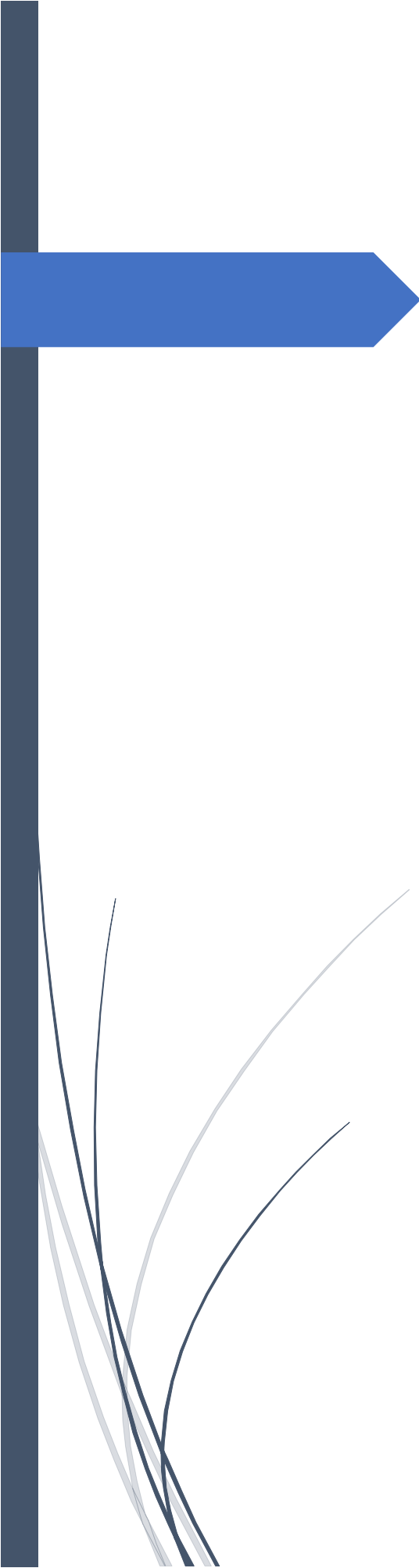




A Deep Dive into Supplements Adverse Events

By Ali Albinfalah

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1. Introduction

The United States Food and Drug Administration (FDA) has been collecting adverse event reports related to food and dietary supplements since the late 1990s through a system now known as the Human Foods Complaint System (HFCS), previously called the CFSAN Adverse Event Reporting System (CAERS). This system serves as the FDA's primary mechanism for monitoring the safety of food products, dietary supplements, and cosmetics after they have reached consumers.

Adverse events are submitted through the FDA's MedWatch reporting program using Form 3500A for mandatory reports and Form 3500B for voluntary consumer submissions. In its early years, reports were submitted primarily by mail or fax, limiting both the volume and accessibility of the reporting system. The introduction of electronic submission and the launch of the consumer-friendly Form 3500B in 2013 significantly lowered the barrier to reporting, making the system more accessible to everyday consumers and healthcare providers.

The HFCS dataset used in this analysis contains adverse event reports from 1999 to 2025, covering more than 40 product categories including dietary supplements and food products. Cosmetic product reports were excluded from this analysis to maintain a focus on food and dietary supplement safety. Reports are submitted by a mix of sources: manufacturers fulfilling mandatory obligations, and consumers, caregivers, and healthcare providers submitting voluntarily. Each report captures information about the product involved, the adverse outcomes experienced, patient demographics where available, and the symptoms reported using standardized MedDRA terminology.

2. Problem Statement

Between 1999 and 2025, the FDA recorded over 90,000 adverse event reports linked to food and dietary supplements. Unlike pharmaceuticals, supplements do not require FDA approval before reaching consumers making post market reporting the only safety net.

3. Objectives

- How have reports changed from 1999 to 2025?
- Which product categories generate the most reports?
- Does gender and age influence supplement reporting?
- What symptoms do supplement users most commonly report?
- Which supplements are linked to most deaths?

4. Target Audience

This analysis is intended for healthcare professionals seeking data driven insights to guide patient counseling on dietary supplement safety, adverse event risks, and demographic vulnerability patterns. It is also relevant to regulatory professionals monitoring post market adverse event trends to inform policy decisions and enforcement priorities.

5. Data Info

- **Shape of the Dataset Before Cleaning:** 231,897 rows and 12 columns
- **Shape of the Dataset After Cleaning:** 113,081 rows and 17 columns
- **Dataset Source:** FDA Human Foods Complaint System (HFCS)

Data Dictionary (After Cleaning):

Column	Type	Description
DATE_REPORTED	datetime	The date on which the case data was first received by the FDA.
REPORT_ID	str	The unique number that identifies each case.
PRODUCT	str	The verbatim brands and/or product names indicated to have been used by the consumer reported to have experienced the adverse event. An adverse event report may specify consumption of a single product or multiple products.
PRODUCT_CATEGORY	category	The FDA industry description associated with the type of product reported. (ice cream products, cosmetics, coffee/tea)
SEX	category	The sex of the individual reported to have experienced the adverse event.
SYMPTOMS	str	The symptom(s) experienced by the injured consumer as specified by the reporter and coded by FDA according to the Medical Data Dictionary for Regulatory Activities (MedDRA). ¹ Each adverse event report may indicate one or more symptoms.
CASE_OUTCOME	str	Outcome(s) of the adverse event experienced by the injured consumer as specified by the reporter; each report may indicate one or more outcomes.
AGE_YEARS	float	Patient age standardized to years. Converted from original AGE and AGE_UNITS columns which recorded age in mixed units (years, months, days, weeks, decades). Null where age was not reported.

AGE_GROUP	category	Patient age bucketed into five groups: Minors (Under 18), Young Adults (18–34), Middle Aged Adults (35–54), Older Adults (55–64), and Seniors (65+). Null ages labeled as Not Reported.
DEATH	int	Indicates whether death was reported as an outcome. 1 = Death reported, 0 = Not reported. Derived from CASE_OUTCOME via keyword matching.
LIFE_THREATENING	int	Indicates whether a life-threatening event was reported. 1 = Life threatening reported, 0 = Not reported. Derived from CASE_OUTCOME via keyword matching.
HOSPITALIZATION	int	Indicates whether hospitalization was reported as an outcome. 1 = Hospitalization reported, 0 = Not reported. Derived from CASE_OUTCOME via keyword matching.
DISABILITY	int	Indicates whether disability or permanent damage was reported as an outcome. 1 = Disability reported, 0 = Not reported. Derived from CASE_OUTCOME via keyword matching.
CONGENITAL_ANAMOLY	int	Indicates whether a congenital anomaly or birth defect was reported. 1 = Reported, 0 = Not reported. Derived from CASE_OUTCOME via keyword matching.
ALLERGIC_REACTION	int	Indicates whether an allergic reaction was reported as an outcome. 1 = Reported, 0 = Not reported. Derived from CASE_OUTCOME via keyword matching.
INJURY	int	Indicates whether a physical injury was reported as an outcome. 1 = Reported, 0 = Not reported. Derived from CASE_OUTCOME via keyword matching.
YEAR	int	Year extracted from DATE_REPORTED

NOTE: For more details on the data info, please refer to the Jupyter Notebook.

6. Data Handling

Removing Concomitant Products

The dataset contains two product type classifications:

- **SUSPECT (84.68%, 196,370 rows)** - Products the reporter believed caused the adverse event
- **CONCOMITANT (15.32%, 35,527 rows)** - Other products the patient was taking at the same time

- **CONCOMITANT** rows represent products taken alongside the suspected product but not believed to have caused the adverse event. Including them would conflate background context with causal association. They also contain significantly higher null rates across key analytical columns, so their rows were removed to keep the analysis focused on products directly associated with harm.

Removing Cosmetic Products

- Cosmetic products were removed despite being the largest category due to the focus of our analysis only being on food and supplements

Handling Duplicates

- There were a total of 421 duplicated rows, and they were dropped.
- There are 16664 duplicates in **REPORT_ID**, and that's expected because one report can involve multiple products. So, each row gets its own row but shares the same report id. For example, a person takes 3 supplements and reports an adverse event, that's one report id with 3 rows, one per product. Those duplicates were not dropped.

Renaming Columns

- Renamed **DATE_FDA_FIRST_RECEIVED_REPORT** to **DATE_REPORTED**
- Renamed **DESCRIPTION** to **PRODUCT_CATEGORY**
- Renamed **PATIENT_AGE** to **AGE**
- Renamed **CASE_MEDDRA_PREFERRED_TERMS** to **SYMPTOMS**

Removing Extra Spaces

- Removes extra spaces from the string columns using strip.

Changing Column Types

- Changed **DATE_REPORTED** type to datetime
- Changed **DATE_EVENT** type to datetime
- Changed **PRODUCT_CATEGORY** to category
- Changed **SEX** type to category

Handling Nulls

The table below shows which columns have nulls and what actions will be taken

Column	Null Count	Null Percentage	Action
DATE_EVENT	42,773	37.83%	Nulls will be left as they are because it's a date column and shouldn't contain text placeholders.
AGE	44,622	39.46%	Will be handled later during age group bucketing
AGE_UNITS	44,622	39.46%	Will be handled later during age group bucketing
SEX	10,162	8.99%	Will be filled with "Not Reported"

Age Standardization and Grouping

1. Replaced unrealistic age values - any age above 105 years was replaced with null, as values exceeding this threshold are likely data entry errors
2. Created **AGE_YEARS** column - a copy of **AGE** used as the base for unit conversion
3. Standardized age units to years - **AGE_UNITS** contained mixed units (months, days, weeks, decades). Each was converted to years using the appropriate formula:
 - Months ÷ 12
 - Days ÷ 365
 - Weeks ÷ 52
 - Decades × 10
4. Bucketed ages into 5 groups - a function was applied to **AGE_YEARS** to assign each patient to an age group. Null ages were labeled "Not Reported".

Age Group	Range
Minors	Under 18
Young Adults	18 - 34
Middle Aged Adults	35 - 54
Older Adults	55 - 64
Seniors	65+

5. Converted **AGE_GROUP** to category type to reduce memory usage and improves performance.

Outcome Categorization

The **CASE_OUTCOME** column contains 386 unique combinations of outcomes since reporters can select multiple outcomes per case. Rather than analyzing every combination, a binary column (1 = present, 0 = absent) was created for each clinically meaningful outcome using keyword matching on the **CASE_OUTCOME** column. This approach preserves the full detail of each report without forcing ambiguous combinations into predefined severity tiers.

The new created columns : **DEATH, LIFE_THREATENING, HOSPITALIZATION, DISABILITY, CONGENITAL_ANAMOLY, ALLERGIC_REACTION, INJURY**

Dropping Unnecessary Columns

The table below shows which columns will be dropped and why

Column	Reason for Dropping
AGE	Replaced by AGE_YEARS and AGE_GROUP
AGE_UNITS	Will No longer be needed since all ages were converted to years
PRODUCT_CODE	The FDA industry code associated with the type of product reported. Will not be used in analysis
DATE_EVENT	Has nulls and the dates may not be accurate. DATE_REPORTED will used for trend analysis instead

NOTE: For more details on the data handling, please refer to the Jupyter Notebook.

7. Analysis

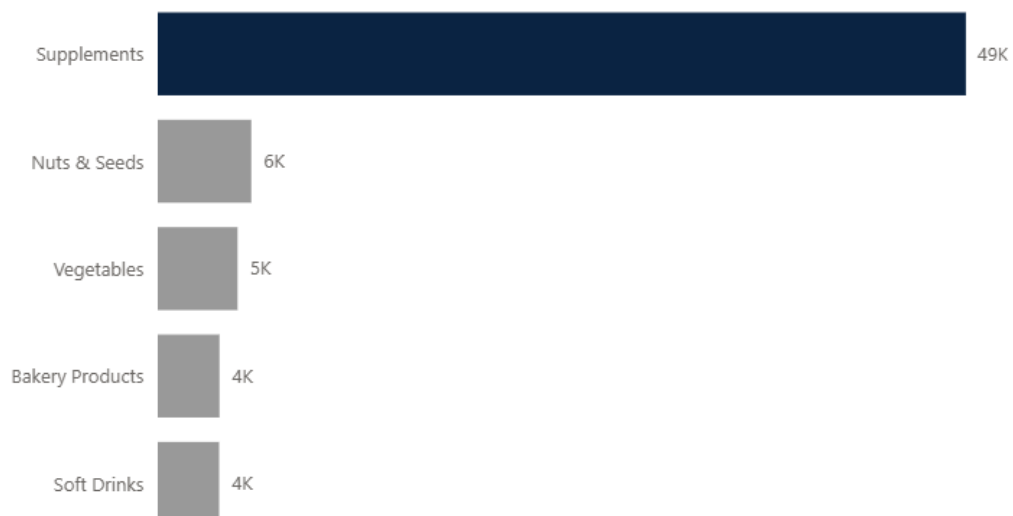
How have reports changed from 1999 to 2025?



Key Insights:

- Adverse event reports were low before 2004, largely due to the reliance on paper forms and fax submissions.
- Growth accelerated following the 2006 Dietary Supplement and Nonprescription Drug Consumer Protection Act (implemented in Dec 2007), which made it mandatory for supplement manufacturers to report serious adverse events.
- Increasing digital accessibility and growing public awareness of the reporting system further contributed to the steady rise in submissions over time.

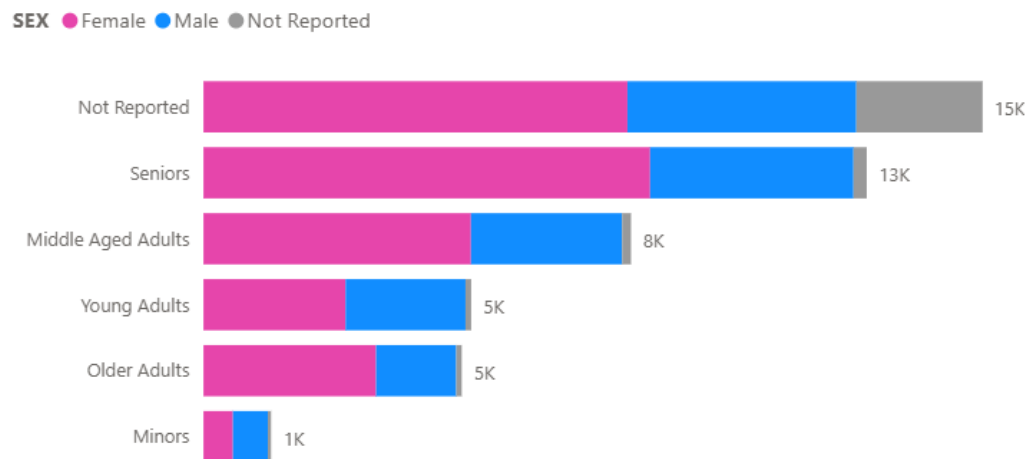
Which product categories have the most reports?



Key Insights:

- Supplements account for the majority of adverse event reports. This large disparity is explained by the Dietary Supplement and Nonprescription Drug Consumer Protection Act, which legally requires supplement manufacturers to report serious adverse events to the FDA, an obligation that does not apply to conventional food producers.
- The remaining four categories each account for under 6,000 reports, primarily driven by allergen reactions, contamination incidents, and additive-related complaints.

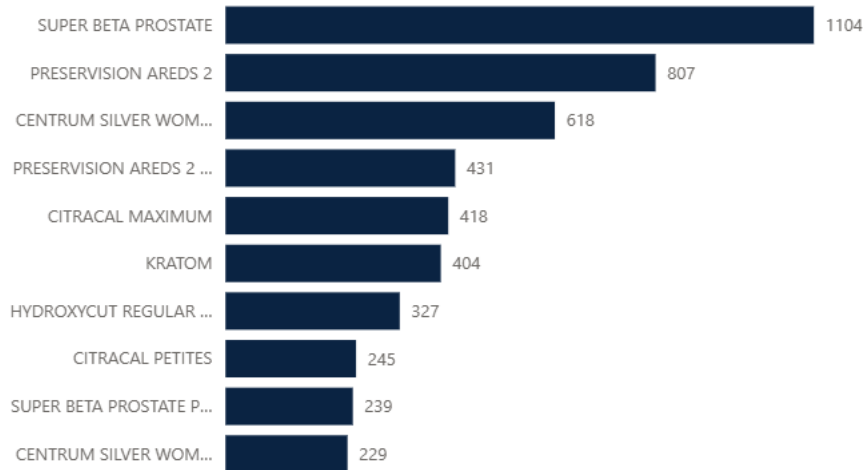
Does gender and age influence supplement reporting?



Key Insights:

- Not Reported (15K) is the largest single group - age and sex are optional fields in the FDA MedWatch reporting form, meaning nearly 40% of reporters did not disclose their age.
- Females outnumber males across all age groups - reflecting higher supplement consumption among women, who are the primary target market for multivitamins, calcium, and weight loss supplements, as well as a well documented tendency among women to report adverse health experiences more frequently than men.
- Seniors (65+) are the most affected age group - older adults managing chronic conditions are more likely to take multiple supplements and medications simultaneously, increasing the risk of adverse interactions.
- Middle Aged Adults (35–54) ranks second - a demographic with high supplement usage for weight management, energy, and preventive health.

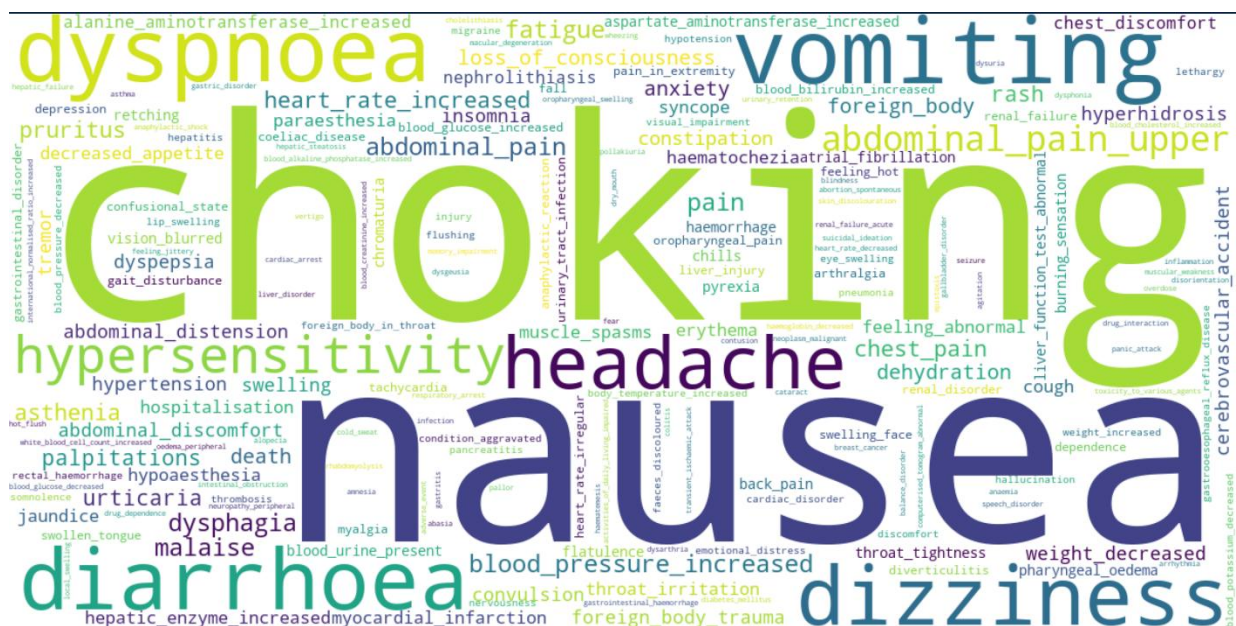
What Are the Most Reported Supplement Products?



Key Insights:

- Super Beta Prostate leads with 1,104 reports - a prostate health supplement widely marketed to older men. Its high report volume reflects widespread use in an aging demographic.
- PreserVision AREDS 2 appears twice - the regular formula and soft gel version combined account for over 1,200 reports, making it the most reported eye health supplement. It is primarily used by elderly patients managing macular degeneration.
- Citracal appears twice as well - Maximum and Petites versions combined, both associated primarily with choking complaints given their large calcium tablet form.

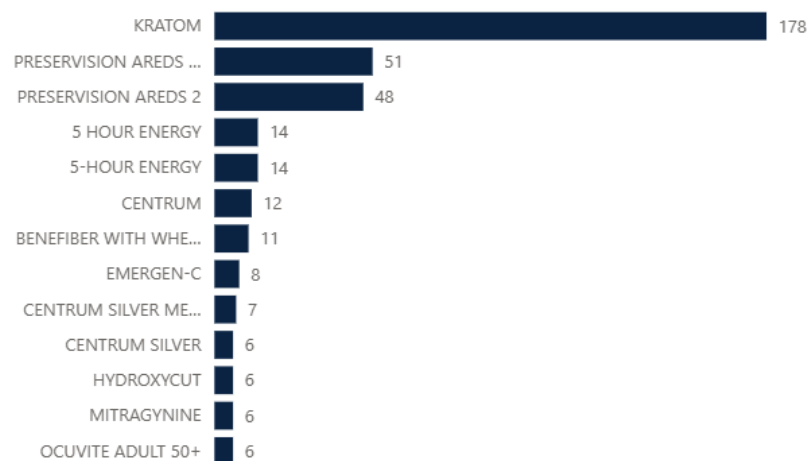
What are the most frequent symptoms in supplements reports?



What are the supplements associated with the common symptoms?

Symptom	Associated Supplements
Choking	Large multivitamins, fish oil softgels, calcium tablets
Nausea / Vomiting	Iron, Zinc (on empty stomach), high dose Vitamin C
Diarrhea	Magnesium, high dose Vitamin C, fiber supplements
Dizziness	Preworkouts, weight loss supplements (Hydroxycut)
Headache	High dose Vitamin D, weight loss supplements (Hydroxycut), preworkouts

Which supplements are linked to most deaths?



- Kratom dominates with 178 deaths - confirming it as the most dangerous supplement in the dataset. The FDA has issued multiple safety warnings about kratom's opioid like effects.
- 5-Hour Energy appears twice - as separate entries due to inconsistent naming, combined they account for 28 deaths, linked to cardiovascular adverse events from high caffeine content.
- PreserVision accounts for 99 combined deaths - surprisingly high for an eye supplement, this likely reflects the elderly demographic who use it rather than the product's inherent toxicity. Macular degeneration patients are typically older adults with multiple comorbidities, making death a correlated outcome rather than a caused one.
- Benefiber, Emergen-C, and Centrum appearing in the deaths chart should be interpreted with caution, these are widely consumed, generally considered safe supplements. Their presence here likely reflects the same confounding pattern, older

or chronically ill users who happened to be taking these products at the time of death rather than products that caused harm.

NOTE: For more detailed information about the analysis, please refer to the POWER BI file.

8. Summary

- Supplements account for 51% of all adverse event reports.
- Females and Seniors (65+) are the most affected demographic.
- Choking and gastrointestinal symptoms are the most reported adverse effects.
- Kratom is associated with the most reported deaths.

9. Recommendations

- Screen patients for herbal supplement use, particularly those on prescription medications where interaction risk is highest.
- Prioritize supplement safety discussions with seniors, patients aged 65+ account for the highest adverse event volumes.
- Address choking risk with elderly patients, recommend smaller tablet formats or powder alternatives for patients with swallowing difficulties.
- Use HFCS as a clinical reference, a valuable and underutilized resource for identifying emerging supplement safety signals before they become widespread.

10. Limitations

- Reports are self reported and not confirmed proof that the product caused the event.
- Side effects may be linked to other health conditions.
- Report quality varies, some reports are incomplete or missing key details.
- Incomplete contact information limits FDA's ability to fully investigate reported events.

11. References

- <https://www.fda.gov/food/compliance-enforcement-food/human-foods-complaint-system-hfcs>
- <https://www.fda.gov/safety/reporting-serious-problems-fda/what-serious-adverse-event>
- <https://www.fda.gov/food/dietary-supplements>
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- <https://www.mayoclinic.org/diseases-conditions/prescription-drug-abuse/in-depth/kratom/art-20402171>
- <https://www.fda.gov/news-events/public-health-focus/fda-and-kratom>
- <https://www.fda.gov/food/information-select-dietary-supplement-ingredients-and-other-substances/fda-warns-consumers-about-pure-and-highly-concentrated-caffeine>